

Will Bitcoin Rule the Free World?

Bitcoin is digital currency that uses peer-to-peer technology to enable instant payments to anyone. It uses open source software called Bitcoin Core to enable the use of the currency.

Every once in a while, a new idea is born—one that has the potential to question the very foundations of everything we know about the world as it is today. The ideas I am talking about are those that have changed and will change the world for the better. Take email, for example—at its inception, people thought it was for the geeks and wouldn't catch on. I would have agreed in the early days, but look at how the world has become so dependent on it over the last few decades. This is an ongoing phenomenon—what is old will either perish or evolve. The only constant, if any, is change.

On a cold winter morning of January 2009, Satoshi Nakamoto set out to change the way we transact business and unleashed the world's first cryptocurrency, bitcoin.

A cryptocurrency is a form of digital currency that uses cryptography (the science of hiding messages in codes that appear to be a random string of characters) as a means of generating new units and securing transactions. Its heavy dependence on cryptography is how it derived its name.

Though Bitcoin is by far the most popular cryptocurrency, a few others like Litecoin and Ripple are also gaining momentum.

The history of Bitcoin

In 2008, Satoshi Nakamoto, founder of Bitcoin, published a paper titled *Bitcoin: A Peer-to-Peer Electronic Cash System* (<http://goo.gl/MqyCW7>), which explained in detail how a peer-to-peer (P2P) currency would work. In January 2009, he officially launched the Bitcoin network. Complementing that, he also launched the Bitcoin Core software (<http://goo.gl/gUbUq>), which would help people mine (a term explained later) and trade (please refer to *An Introduction to Bitcoin: The Open Source Cryptographic Currency*, published in the November 2013 edition of *Open Source For You*) in bitcoins. The Bitcoin Project was released under the MIT licence, which essentially meant that it's free and open source. The new currency rapidly gained momentum and became the talk of the town. It proved to be a nightmare for banks and governments worldwide but it became the darling of the online underworld (for reasons that I will go into later).



After launching Bitcoin, Satoshi Nakamoto gradually disappeared from the online world leaving his creation in the hands of the community, which was by then swearing by it. Surprisingly, in this well-connected and spied-upon world, Nakamoto still remains a mysterious figure. While many have speculated about who he really is, no one has arrived at any conclusive proof about his identity. Given the complexity and sophistication of the Bitcoin system, many researchers believe that it can't be the work of a single person. They believe that some kind of group or consortium of technology companies is behind this cryptocurrency. Considering that Bitcoin mining is processing power-intensive, these companies would surely stand to benefit from selling the concept to the Bitcoin community.

What is a bitcoin?

Given its name and logo, people mistakenly assume that it is a physical coin with electromagnetic capabilities, which is contrary to the very notion of a cryptocurrency. The essence of a cryptocurrency is that it is not required to and cannot exist in a physical form.

The following analogy will help you understand the concept of Bitcoin better. Let's say you have 10,000 units of a currency in a bank account and you can conduct online transactions with this account. At any given moment,